The FREUID Challenge 2026

Technical Report — Marc Donovici

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Abstract

We address binary identity-document fraud detection under cross-document, digital, and print–capture domain shift. Our frozen submission is a no-TTA weighted rank ensemble of a ConvNeXt-V2 Base detector (weight 0.75) and a DINOv2 ViT-B/14-with-registers detector (weight 0.25). The former supplies the strongest observed physical-recapture operating point; the latter adds representations trained across unseen layouts and attack families. Recapture simulation is applied to both classes, preventing capture appearance from becoming a fraud shortcut. Model selection uses only official training data and a disjoint MIDV-Holo probe; private test images and labels are never used for fitting or selection. A one-shot three-member training run was completed before private release, but two auxiliary models collapsed on the unseen-attack probe and were excluded. Per-template score normalization was also rejected after worsening the public FREUID Score from 0.33816 to 0.54141 (lower is better). The public repository contains pinned weights, source, an immutable-base Dockerfile, and an offline entrypoint that maps flat filenames in `/data` to exactly `/submissions/submission.csv`.

Competition: FREUID Challenge 2026 (IJCAI–ECAI 2026)
Kaggle team: marcdonovici
Code: https://github.com/Marc-Dvci/FREUID, frozen source commit `4db0ae5fd62c739a9398175c78e32add4f418a22`

1 Problem and metric

FREUID targets bona-fide versus fraudulent identity documents, including physical manipulation, generative editing, and print–capture attacks on document domains under-represented in public benchmarks [2]. Labels are 0 for bona-fide and 1 for fraud; larger submitted scores mean more likely fraudulent. The minimized FREUID Score is the harmonic combination of goodness terms $1 - \text{AuDET}$ and $1 - \text{APCER}_{\text{BPCER}=1\%}$. Consequently, a small tail of high-scored bona-fides can be more damaging than average classification error suggests.

2 Method

2.1 Selected detectors

The selected ConvNeXt-V2 Base checkpoint uses the FCMAE ImageNet-22k/1k initialization [4], a linear binary head, and 512×512 input (87,693,825 parameters). It was fine-tuned with a leave-one-document-domain-out split, medium-strength recapture augmentation, and gradient checkpointing during training. The selected DINOv2 model uses ViT-B/14 with register tokens and LVD-142M initialization [1], a linear head, and 392×392 input (86,133,505 parameters). Both produce one logit and use sigmoid probabilities before ensembling.

For a test set of n images, each detector's probabilities are mapped to average ranks $r_i \in [0, 1]$; tied probabilities receive the same average rank. The final fraud score is

$$s_i = 0.75\, r_i^{\text{ConvNeXt}} + 0.25\, r_i^{\text{DINOv2}}. \tag{1}$$

These weights, checkpoints, and no-TTA policy were frozen before private-test release. Rank combination controls incompatible calibration and limits influence from the less physically robust DINOv2 member. Horizontal-flip TTA was rejected because it worsened DINOv2 on the full unseen probe (FREUID 0.97752 to 0.99371).

2.2 Training

Training uses binary cross-entropy with logits, AdamW, cosine decay after a one-epoch warm-up, automatic mixed precision, channels-last tensors, and seed 42. DINOv2 was fine-tuned for three epochs with batch size 16, two-step gradient accumulation (effective batch 32), learning rate 10^{-4}, weight decay 0.01, and recapture probability 0.35. Epoch 2 was selected by the disjoint OOD probe (checkpoint selection score 0.97484 on its fixed 3,000-row cap). The ConvNeXt checkpoint used six epochs, batch size 8, four-step accumulation, learning rate 10^{-4}, weight decay 0.01, dropout 0.2, and recapture probability 0.6; its full arguments are embedded in the checkpoint.

The recapture transform models resize/upsample, repeated JPEG compression, blur, moiré, halftoning, glare, vignetting, paper grain, colour cast, and sensor noise. It is sampled independently for both classes. A pre-freeze one-shot run additionally trained ConvNeXt-V2 Base at 384 pixels and a constrained Bayar/SRM ConvNeXt-V2 Nano forensic model. Their OOD outputs were near-constant; they were excluded rather than allowed to create arbitrary filename-order ranks.

3 Data and validation

The official training release contains 69,352 samples from five document types (40,005 bona-fide, 29,347 fraud). Competition images are not redistributed. The DINOv2 run holds the 20 official real-recapture rows out of fitting, reserves a fixed stratified 5% monitoring split, and adds 15,179 MIDV-Holo frames to training (9,070 bona-fide, 6,109 fraud) [3]. The external training partition covers 14 documents and two attack families. A completely disjoint probe covers 6 other documents and 2 unseen attack families (6,428 rows); there is no shared image, clip, or document identity. MIDV-Holo is CC BY-SA 2.5 and includes Generated Photos identities as documented by its authors. Pretrained DINOv2 parameters are Apache-2.0; ConvNeXt-V2 pretrained parameters are CC BY-NC 4.0. Full notices and URLs are in `THIRD_PARTY_NOTICES.md`.

The small 20-row official real-recapture slice is reported only as a stress test. Public leaderboard feedback is a secondary sanity check because its domains and prevalence need not represent the 134,997-row private release. No public or private test image participates in training, adaptation, clustering, or checkpoint selection.

4 Results and ablations

Table 1: Pre-freeze evaluation (lower is better). The OOD probe uses all 6,428 rows.

Candidate / split	FREUID	AuDET	APCER@1% BPCER
DINOv2, unseen MIDV-Holo	0.97752	0.50052	0.98850
ConvNeXt, unseen MIDV-Holo	0.98288	0.54507	0.99127
Frozen 0.75/0.25 ensemble, unseen MIDV-Holo	0.98672	0.54191	0.99326
ConvNeXt, official real-recapture stress	0.10584	0.06548	0.14286
DINOv2, official real-recapture stress	0.67570	0.40476	0.77714
Frozen 0.75/0.25 ensemble, real-recapture stress	0.14890	0.08333	0.20571

The conflicting probes motivate conservative weighting: DINOv2 is marginally best on new external documents, whereas ConvNeXt is decisively better on the only official physical-recapture sample. The existing ConvNeXt public submission scored 0.33816. Per-template clustering and within-template normalization degraded it to 0.54141 and is disabled. The exact frozen ensemble public score remains **pending Kaggle processing at report build time**; it must be filled before the final PDF.

5 Inference and Docker reproduction

Inference resizes and normalizes each image according to its checkpoint, uses FP16 autocast on CUDA, and does not use TTA. The Docker entrypoint scans only supported image files directly below `/data`; the identifier is the filename stem. It rejects empty input, duplicate stems, missing weights, invalid ensemble weights, non-finite predictions, or any output path other than `/submissions/submission.csv`. The output has exactly `id,label`; all labels are finite in $[0, 1]$. The root filesystem can be read-only, model creation uses `pretrained=False`, CUDA kernel caching is disabled, and no runtime network or model cache is needed.

```
git lfs install && git lfs pull
docker build -t freuid-repro:local .
```

```
docker run --rm --gpus all --network none --read-only \
  -v /path/to/flat/private_images:/data:ro \
  -v "$(pwd)/out:/submissions" freuid-repro:local
```

The PyTorch base is pinned to the following digest:

`sha256:77f17f843507062875ce8be2a6f76aa6`
`aa3df7f9ef1e31d9d7432f4b0f563dee`

Python dependencies are fully version-pinned. The private test set was scored by running this container unchanged on the 134,997 released private images, under `-network none`, `-read-only`, a read-only `/data`, and a writable `/submissions`. The measured wall-clock time was 131 minutes on one RTX 4070 (both members sequentially, batch 64, zero dataloader workers), which is well below the organizer's six-hour A100 limit. Throughput is bound by single-process image decoding rather than by the GPU, so a host with more dataloader workers will be faster. Hashes and the image digest are recorded in `FINAL_AUDIT.md`.

The selected public-row CSV retains placeholder 0.5 values for unreleased IDs. After private release, the unchanged container runs on private images alone. `scripts/assemble_final_submission.py` requires those IDs to equal exactly the official sample IDs minus the public-image stems, then replaces only untouched private placeholders. It preserves official row order and validates the final hash, schema, range, and finiteness.

6 Reproducibility and integrity

Training used one NVIDIA RTX 4070 (12 GB), PyTorch 2.6.0, CUDA 12.4, and seed 42. The full one-shot three-model run took 292.8 minutes. Checkpoint SHA-256 values are:

- ConvNeXt selected: `aebf36acdf23dfe7a1542e9b07ba94e782910b547d6244f3fbb8d84083066510`.
- DINOv2 selected: `f333c566bef9d6c2e9be8ba4a5b5efb5e708492e71162a135856a5e08ee1e8ca`.

cuDNN benchmarking can select nondeterministic convolution kernels, so bitwise retraining is not claimed; submitted weights and inference are fixed. The repository contains no competition images, external images, credentials, or runtime download logic.

7 Team and contributions

Marc Donovici performed data preparation, model development, validation, container engineering, and reporting, and is the sole entrant.

References

[1] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[2] Ivan Relić et al. The FREUID challenge 2026: 1st competition on identity documents fraud detection. https://freuid2026.microblink.com/, 2026.

[3] Alexander Sheshkus, Vladimir V. Arlazarov, Dmitry V. Polevoy, et al. MIDV-Holo: A new dataset for identity document analysis and hologram detection. In *Document Analysis and Recognition – ICDAR 2023 Workshops*, pages 219–232, 2023.

[4] Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. Convnext v2: Co-designing and scaling convnets with masked autoencoders. In *CVPR*, 2023.